

DEPLOYING MACHINE LEARNING MODELS IN PRODUCTION

03 ungraded lab introduction to docker

C4 W1 Lab 1 Docker Intro

Source: 03_ungraded-lab-introduction-to-docker_C4_W1_Lab_1_Docker_Intro.md

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property="og:image:width" content="1200" /><meta property="og:image:height" content="600"
/><meta property="og:site_name" content="GitHub" /><meta property="og:type"
content="object" /><meta property="og:title" content="machine-learning-engineering-for-
production-public/course4/week1-ungraded-labs/C4_W1_Lab_1_Docker_Intro.md at main · https-
deeplearning-ai/machine-learning-engineering-for-production-public" /><meta
property="og:url" content="https://github.com/https-deeplearning-ai/machine-learning-
engineering-for-production-public" /><meta property="og:description" content="Public repo
for DeepLearning.AI MLEP Specialization - machine-learning-engineering-for-production-
public/course4/week1-ungraded-labs/C4_W1_Lab_1_Docker_Intro.md at main · https-
deeplearning-ai/mach..." />

  <meta name="hostname" content="github.com">

  <meta name="expected-hostname" content="github.com">

```



```
<meta name="enabled-features"
content="TURBO_EXPERIMENT_RISKY, IMAGE_METRIC_TRACKING, GEOJSON_AZURE_MAPS, SUDO_MODAL_FOCUS">
```

```
<meta data-hydrostats="publish">
```

```
<link rel="canonical" href="https://github.com/https-deeplearning-ai/machine-learning-
engineering-for-production-public/blob/main/course4/week1-ungraded-
labs/C4_W1_Lab_1_Docker_Intro.md" data-turbo-transient>
```

```
<div class="position-relative js-header-wrapper ">
  <a href="#start-of-content" class="px-2 py-4 color-bg-accent-emphasis color-fg-on-
emphasis show-on-focus js-skip-to-content">Skip to content</a>
  <span data-view-component="true" class="progress-pjax-loader Progress position-fixed
width-full">
<span style="width: 0%;" data-view-component="true" class="Progress-item progress-pjax-
loader-bar left-0 top-0 color-bg-accent-emphasis"></span>
```

```
<script crossorigin="anonymous" defer="defer" type="application/javascript"
src="https://github.githubassets.com/assets/vendors-node_modules_github_remote-
form_dist_index_js-node_modules_github_memoize_dist_esm_in-687f35-d131f0b6de8e.js">
</script>
```

Toggle navigation



```

<div class="flex-1">
  <a href="/signup?
ref_cta=Sign+up&ref_loc=header+logged+out&ref_page=%2F%3Cuser-name%3E%2F%3Crepo-
name%3E%2Fblob%2Fshow&source=header-repo"
    class="d-inline-block d-lg-none flex-order-1 f5 no-underline border color-border-
default rounded-2 px-2 py-1 color-fg-inherit"
    data-hydro-click="
{&quot;event_type&quot;:&quot;authentication.click&quot;;&quot;payload&quot;:;
{&quot;location_in_page&quot;:&quot;site
header&quot;;&quot;repository_id&quot;:null,&quot;auth_type&quot;:&quot;SIGN_UP&quot;;&quot;
;originating_url&quot;:&quot;https://github.com/https-deeplearning-ai/machine-learning-
engineering-for-production-public/blob/main/course4/week1-ungraded-
labs/C4_W1_Lab_1_Docker_Intro.md&quot;;&quot;user_id&quot;:null}}" data-hydro-click-
hmac="5ad6742d9419afed9af576678a7ab250a31ed9b65daa021f8cfac1a88e5a4253"
    >
    Sign&nbsp;up
  </a>
</div>

<div class="flex-1 flex-order-2 text-right">
  <button aria-label="Toggle navigation" aria-expanded="false" type="button" data-view-
component="true" class="js-details-target Button--link Button--medium Button d-lg-none
color-fg-inherit p-1">    <span class="Button-content">
    <span class="Button-label"><div class="HeaderMenu-toggle-bar rounded my-1"></div>
    <div class="HeaderMenu-toggle-bar rounded my-1"></div>
    <div class="HeaderMenu-toggle-bar rounded my-1"></div></span>
</span>
</div>

```

```

<div class="HeaderMenu--logged-out p-responsive height-fit position-lg-relative d-lg-flex
flex-column flex-auto pt-7 pb-4 top-0">
  <div class="header-menu-wrapper d-flex flex-column flex-self-end flex-lg-row flex-justify-
between flex-auto p-3 p-lg-0 rounded rounded-lg-0 mt-3 mt-lg-0">
    <nav class="mt-0 px-3 px-lg-0 mb-3 mb-lg-0" aria-label="Global">
      <ul class="d-lg-flex list-style-none">
        <li class="HeaderMenu-item position-relative flex-wrap flex-justify-between flex-
items-center d-block d-lg-flex flex-lg-nowrap flex-lg-items-center js-details-container js-
header-menu-item">
          <button type="button" class="HeaderMenu-link border-0 width-full width-lg-auto px-0 px-
lg-2 py-3 py-lg-2 no-wrap d-flex flex-items-center flex-justify-between js-details-target"
aria-expanded="false">
            Product
            <svg opacity="0.5" aria-hidden="true" height="16" viewBox="0 0 16 16" version="1.1"
width="16" data-view-component="true" class="octicon octicon-chevron-down HeaderMenu-icon
ml-1">
<path d="M12.78 5.22a.749.749 0 0 1 0 1.06l-4.25 4.25a.749.749 0 0 1-1.06 0L3.22
6.28a.749.749 0 1 1 1.06-1.06L8.93 13.72-3.719a.749.749 0 0 1 1.06 0Z"></path>

```



Actions

Automate any workflow



[Packages](#)

[Host and manage packages](#)



[Security](#)

[Find and fix vulnerabilities](#)



[Codespaces](#)

[Instant dev environments](#)



[Copilot](#)

[Write better code with AI](#)



[Code review](#)

[Manage code changes](#)



[Issues](#)

[Plan and track work](#)



Discussions

Collaborate outside of code

```

    </ul>
  </div>
  <div class="px-lg-4">
    <span class="d-block h4 color-fg-default my-1" id="product-explore-
heading">Explore</span>
    <ul class="list-style-none f5" aria-labelledby="product-explore-heading">
      <li>

```

All features

```
<li>
```

Documentation

```

<svg aria-hidden="true" height="16" viewBox="0 0 16 16" version="1.1" width="16" data-
view-component="true" class="octicon octicon-link-external HeaderMenu-external-icon
color-fg-subtle">
<path d="M3.75 2h3.5a.75.75 0 0 1 0 1.5h-3.5a.25.25 0 0 0-.25.25v8.5c0
.138.112.25.25.25h8.5a.25.25 0 0 0 .25-.25v-3.5a.75.75 0 0 1 1.5 0v3.5A1.75 1.75 0 0 1
12.25 14h-8.5A1.75 1.75 0 0 1 2 12.25v-8.5C2 2.784 2.784 2 3.75 2Zm6.854-1h4.146a.25.25
0 0 1 .25.25v4.146a.25.25 0 0 1-.427.177L13.03 4.03 9.28 7.78a.751.751 0 0 1-
1.042-.018.751.751 0 0 1-.018-1.042l3.75-3.75-1.543-1.543A.25.25 0 0 1 10.604 12">
</path>

```

```
<li>
```

GitHub Skills

```

<svg aria-hidden="true" height="16" viewBox="0 0 16 16" version="1.1" width="16" data-
view-component="true" class="octicon octicon-link-external HeaderMenu-external-icon
color-fg-subtle">
<path d="M3.75 2h3.5a.75.75 0 0 1 0 1.5h-3.5a.25.25 0 0 0-.25.25v8.5c0
.138.112.25.25.25h8.5a.25.25 0 0 0 .25-.25v-3.5a.75.75 0 0 1 1.5 0v3.5A1.75 1.75 0 0 1
12.25 14h-8.5A1.75 1.75 0 0 1 2 12.25v-8.5C2 2.784 2.784 2 3.75 2Zm6.854-1h4.146a.25.25
0 0 1 .25.25v4.146a.25.25 0 0 1-.427.177L13.03 4.03 9.28 7.78a.751.751 0 0 1-
1.042-.018.751.751 0 0 1-.018-1.042l3.75-3.75-1.543-1.543A.25.25 0 0 1 10.604 12">
</path>

```

```
<li>
```

[Blog](#)

```
<svg aria-hidden="true" height="16" viewBox="0 0 16 16" version="1.1" width="16" data-view-component="true" class="octicon octicon-link-external HeaderMenu-external-icon color-fg-subtle">
<path d="M3.75 2h3.5a.75.75 0 0 1 0 1.5h-3.5a.25.25 0 0 0-.25.25v8.5c0
.138.112.25.25.25h8.5a.25.25 0 0 0 .25-.25v-3.5a.75.75 0 0 1 1.5 0v3.5A1.75 1.75 0 0 1
12.25 14h-8.5A1.75 1.75 0 0 1 2 12.25v-8.5C2 2.784 2.784 2 3.75 2Zm6.854-1h4.146a.25.25
0 0 1 .25.25v4.146a.25.25 0 0 1-.427.177L13.03 4.03 9.28 7.78a.751.751 0 0 1-
1.042-.018.751.751 0 0 1-.018-1.042l3.75-3.75-1.543-1.543A.25.25 0 0 1 10.604 12">
</path>
```

```
</ul>
</div>
</div>
```

```
<li class="HeaderMenu-item position-relative flex-wrap flex-justify-between
flex-items-center d-block d-lg-flex flex-lg-nowrap flex-lg-items-center js-details-container
js-header-menu-item">
  <button type="button" class="HeaderMenu-link border-0 width-full width-lg-auto px-0
px-lg-2 py-3 py-lg-2 no-wrap d-flex flex-items-center flex-justify-between js-details-
target" aria-expanded="false">
    Solutions
    <svg opacity="0.5" aria-hidden="true" height="16" viewBox="0 0 16 16" version="1.1"
width="16" data-view-component="true" class="octicon octicon-chevron-down HeaderMenu-
icon ml-1">
<path d="M12.78 5.22a.749.749 0 0 1 0 1.06l-4.25 4.25a.749.749 0 0 1-1.06 0L3.22
6.28a.749.749 0 1 1 1.06-1.06L8 8.939l3.72-3.719a.749.749 0 0 1 1.06 0Z"></path>
```

For

- [Enterprise](#)

```
<li>
```

[Teams](#)

```
<li>
```

[Startups](#)

```
<li>
```

Education

```
<svg aria-hidden="true" height="16" viewBox="0 0 16 16" version="1.1" width="16"
data-view-component="true" class="octicon octicon-link-external HeaderMenu-external-
icon color-fg-subtle">
<path d="M3.75 2h3.5a.75.75 0 0 1 0 1.5h-3.5a.25.25 0 0 0-.25.25v8.5c0
.138.112.25.25.25h8.5a.25.25 0 0 0 .25-.25v-3.5a.75.75 0 0 1 1.5 0v3.5A1.75 1.75 0 0
1 12.25 14h-8.5A1.75 1.75 0 0 1 2 12.25v-8.5C2 2.784 2.784 2 3.75 2Zm6.854-
1h4.146a.25.25 0 0 1 .25.25v4.146a.25.25 0 0 1-.427.177L13.03 4.03 9.28 7.78a.751.751
0 0 1-1.042-.018.751.751 0 0 1-.018-1.042l3.75-3.75-1.543-1.543A.25.25 0 0 1 10.604
1Z"></path>
```

```
</ul>
</div>
<div class="border-bottom pb-3 mb-3">
  <span class="d-block h4 color-fg-default my-1" id="solutions-by-solution-
heading">By Solution</span>
  <ul class="list-style-none f5" aria-labelledby="solutions-by-solution-
heading">
    <li>
```

CI/CD & Automation

```
<li>
```

DevOps

```
<svg aria-hidden="true" height="16" viewBox="0 0 16 16" version="1.1" width="16"
data-view-component="true" class="octicon octicon-link-external HeaderMenu-external-
icon color-fg-subtle">
<path d="M3.75 2h3.5a.75.75 0 0 1 0 1.5h-3.5a.25.25 0 0 0-.25.25v8.5c0
.138.112.25.25.25h8.5a.25.25 0 0 0 .25-.25v-3.5a.75.75 0 0 1 1.5 0v3.5A1.75 1.75 0 0
1 12.25 14h-8.5A1.75 1.75 0 0 1 2 12.25v-8.5C2 2.784 2.784 2 3.75 2Zm6.854-
1h4.146a.25.25 0 0 1 .25.25v4.146a.25.25 0 0 1-.427.177L13.03 4.03 9.28 7.78a.751.751
0 0 1-1.042-.018.751.751 0 0 1-.018-1.042l3.75-3.75-1.543-1.543A.25.25 0 0 1 10.604
1Z"></path>
```

```
<li>
```

DevSecOps

```
<svg aria-hidden="true" height="16" viewBox="0 0 16 16" version="1.1" width="16"
data-view-component="true" class="octicon octicon-link-external HeaderMenu-external-
icon color-fg-subtle">
<path d="M3.75 2h3.5a.75.75 0 0 1 0 1.5h-3.5a.25.25 0 0 0-.25.25v8.5c0
.138.112.25.25.25h8.5a.25.25 0 0 0 .25-.25v-3.5a.75.75 0 0 1 1.5 0v3.5A1.75 1.75 0 0
1 12.25 14h-8.5A1.75 1.75 0 0 1 2 12.25v-8.5C2 2.784 2.784 2 3.75 2Zm6.854-
1h4.146a.25.25 0 0 1 .25.25v4.146a.25.25 0 0 1-.427.177L13.03 4.03 9.28 7.78a.751.751
0 0 1-1.042-.018.751.751 0 0 1-.018-1.042l3.75-3.75-1.543-1.543A.25.25 0 0 1 10.604
1Z"></path>
```

```
</ul>
</div>
<div class="">
  <span class="d-block h4 color-fg-default my-1" id="solutions-case-studies-
heading">Case Studies</span>
  <ul class="list-style-none f5" aria-labelledby="solutions-case-studies-
heading">
    <li>
```

Customer Stories

```
<li>
```

Resources

```
<svg aria-hidden="true" height="16" viewBox="0 0 16 16" version="1.1" width="16"
data-view-component="true" class="octicon octicon-link-external HeaderMenu-external-
icon color-fg-subtle">
<path d="M3.75 2h3.5a.75.75 0 0 1 0 1.5h-3.5a.25.25 0 0 0-.25.25v8.5c0
.138.112.25.25.25h8.5a.25.25 0 0 0 .25-.25v-3.5a.75.75 0 0 1 1.5 0v3.5A1.75 1.75 0 0
1 12.25 14h-8.5A1.75 1.75 0 0 1 2 12.25v-8.5C2 2.784 2.784 2 3.75 2Zm6.854-
1h4.146a.25.25 0 0 1 .25.25v4.146a.25.25 0 0 1-.427.177L13.03 4.03 9.28 7.78a.751.751
0 0 1-1.042-.018.751.751 0 0 1-.018-1.042l3.75-3.75-1.543-1.543A.25.25 0 0 1 10.604
1Z"></path>
```

```
</ul>
</div>
</div>
```

```

<li class="HeaderMenu-item position-relative flex-wrap flex-justify-between
flex-items-center d-block d-lg-flex flex-lg-nowrap flex-lg-items-center js-details-
container js-header-menu-item">
  <button type="button" class="HeaderMenu-link border-0 width-full width-lg-auto px-0
px-lg-2 py-3 py-lg-2 no-wrap d-flex flex-items-center flex-justify-between js-details-
target" aria-expanded="false">
    Open Source
    <svg opacity="0.5" aria-hidden="true" height="16" viewBox="0 0 16 16"
version="1.1" width="16" data-view-component="true" class="octicon octicon-chevron-
down HeaderMenu-icon ml-1">
<path d="M12.78 5.22a.749.749 0 0 1 0 1.06l-4.25 4.25a.749.749 0 0 1-1.06 0L3.22
6.28a.749.749 0 1 1 1.06-1.06L8.939 3.72-3.719a.749.749 0 0 1 1.06 0Z"></path>

```

```

■ <div>
  <div class="color-fg-default h4">GitHub Sponsors</div>
  Fund open source developers
</div>

```

```

</ul>
</div>
<div class="border-bottom pb-3 mb-3">
  <ul class="list-style-none f5" >
    <li>

```

```

<div>
  <div class="color-fg-default h4">The ReadME Project</div>
  GitHub community articles
</div>

```

```

</ul>
</div>
<div class="">
  <span class="d-block h4 color-fg-default my-1" id="open-source-
repositories-heading">Repositories</span>
  <ul class="list-style-none f5" aria-labelledby="open-source-repositories-
heading">
    <li>

```

Topics

```
<li>
```

Trending

```
<li>
```

Collections

```

    </ul>
  </div>
</div>

```

```

    <li class="HeaderMenu-item position-relative flex-wrap flex-justify-
between flex-items-center d-block d-lg-flex flex-lg-nowrap flex-lg-items-center js-
details-container js-header-menu-item">
<a class="HeaderMenu-link no-underline px-0 px-lg-2 py-3 py-lg-2 d-block d-lg-
inline-block" data-analytics-event="{&quot;category&quot;:&quot;Header menu top
item (logged out)&quot;;&quot;action&quot;:&quot;click to go to
Pricing&quot;;&quot;label&quot;:&quot;ref_cta:Pricing&quot;;}"
href="/pricing">Pricing</a>

```

```

    </ul>
  </nav>

  <div class="d-lg-flex flex-items-center px-3 px-lg-0 mb-3 mb-lg-0 text-center
text-lg-left">
    <div class="d-lg-flex min-width-0 mb-2 mb-lg-0">

```

```

placeholder="Search" data-unscoped-placeholder="Search
GitHub" data-scoped-placeholder="Search" autocapitalize="off" role="combobox" aria-
haspopup="listbox" aria-expanded="false" aria-autocomplete="list" aria-
controls="jump-to-results" aria-label="Search" data-jump-to-suggestions-
path="/_graphql/GetSuggestedNavigationDestinations" spellcheck="false"
autocomplete="off" > <input type="hidden" data-csrf="true" class="js-data-jump-to-
suggestions-path-csrf"
value="mpBzIlnu3Gzg8cc8SGi2wzN3nNPBwoBGlVj67+0rqgiTEg3sjcMTQjbNF6W9x+m170mG+bWVquUG
cEi2WsfJNQ==" /> <input type="hidden" class="js-site-search-type-field" name="type"
> <svg xmlns="http://www.w3.org/2000/svg" width="22" height="20" aria-hidden="true"
class="mr-1 header-search-key-slash"><path fill="none" stroke="#979A9C" opacity=".4"
d="M3.5.5h12c1.7 0 3 1.3 3 3v13c0 1.7-1.3 3-3 3h-12c-1.7 0-3-1.3-3-3v-13c0-1.7 1.3-
3 3-3z"></path><path fill="#979A9C" d="M11.8 6L8 15.1h-.9L10.8 6h1z"></path></svg>
<div class="Box position-absolute overflow-hidden d-none jump-to-suggestions js-
jump-to-suggestions-container">

```




```

<img class="avatar mr-2 flex-shrink-0 js-jump-to-suggestion-avatar d-none"
alt="" aria-label="Team" src="" width="28" height="28">

<div class="jump-to-suggestion-name js-jump-to-suggestion-name flex-auto
overflow-hidden text-left no-wrap css-truncate css-truncate-target">
</div>

<div class="border rounded-2 flex-shrink-0 color-bg-subtle px-1 color-fg-muted
ml-1 f6 d-none js-jump-to-badge-search">
  <span class="js-jump-to-badge-search-text-default d-none" aria-label="in
this repository">
    In this repository
  </span>
  <span class="js-jump-to-badge-search-text-global d-none" aria-label="in all
of GitHub">
    All GitHub
  </span>
  <span aria-hidden="true" class="d-inline-block ml-1 v-align-middle">⌵</span>
</div>

<div aria-hidden="true" class="border rounded-2 flex-shrink-0 color-bg-subtle
px-1 color-fg-muted ml-1 f6 d-none d-on-nav-focus js-jump-to-badge-jump">
  Jump to
  <span class="d-inline-block ml-1 v-align-middle">⌵</span>
</div>

```

- No suggested jump to results
- 📄 🗃️ 🔍

```

<img class="avatar mr-2 flex-shrink-0 js-jump-to-suggestion-avatar d-none"
alt="" aria-label="Team" src="" width="28" height="28">

<div class="jump-to-suggestion-name js-jump-to-suggestion-name flex-auto
overflow-hidden text-left no-wrap css-truncate css-truncate-target">
</div>

<div class="border rounded-2 flex-shrink-0 color-bg-subtle px-1 color-fg-muted
ml-1 f6 d-none js-jump-to-badge-search">
  <span class="js-jump-to-badge-search-text-default d-none" aria-label="in
this repository">
    In this repository
  </span>
  <span class="js-jump-to-badge-search-text-global d-none" aria-label="in all
of GitHub">
    All GitHub
  </span>
  <span aria-hidden="true" class="d-inline-block ml-1 v-align-middle">↵</span>
</div>

<div aria-hidden="true" class="border rounded-2 flex-shrink-0 color-bg-subtle
px-1 color-fg-muted ml-1 f6 d-none d-on-nav-focus js-jump-to-badge-jump">
  Jump to
  <span class="d-inline-block ml-1 v-align-middle">↵</span>
</div>

```



```

<img class="avatar mr-2 flex-shrink-0 js-jump-to-suggestion-avatar d-none"
alt="" aria-label="Team" src="" width="28" height="28">

<div class="jump-to-suggestion-name js-jump-to-suggestion-name flex-auto
overflow-hidden text-left no-wrap css-truncate css-truncate-target">
</div>

<div class="border rounded-2 flex-shrink-0 color-bg-subtle px-1 color-fg-muted
ml-1 f6 d-none js-jump-to-badge-search">
  <span class="js-jump-to-badge-search-text-default d-none" aria-label="in
this organization">
    In this organization
  </span>
  <span class="js-jump-to-badge-search-text-global d-none" aria-label="in all
of GitHub">
    All GitHub
  </span>
  <span aria-hidden="true" class="d-inline-block ml-1 v-align-middle">↵</span>
</div>

<div aria-hidden="true" class="border rounded-2 flex-shrink-0 color-bg-subtle
px-1 color-fg-muted ml-1 f6 d-none d-on-nav-focus js-jump-to-badge-jump">
  Jump to
  <span class="d-inline-block ml-1 v-align-middle">↵</span>
</div>

```



```

<img class="avatar mr-2 flex-shrink-0 js-jump-to-suggestion-avatar d-none"
alt="" aria-label="Team" src="" width="28" height="28">

<div class="jump-to-suggestion-name js-jump-to-suggestion-name flex-auto
overflow-hidden text-left no-wrap css-truncate css-truncate-target">
</div>

<div class="border rounded-2 flex-shrink-0 color-bg-subtle px-1 color-fg-muted
ml-1 f6 d-none js-jump-to-badge-search">
  <span class="js-jump-to-badge-search-text-default d-none" aria-label="in
this repository">
    In this repository
  </span>
  <span class="js-jump-to-badge-search-text-global d-none" aria-label="in all
of GitHub">
    All GitHub
  </span>
  <span aria-hidden="true" class="d-inline-block ml-1 v-align-middle">⌵</span>
</div>

<div aria-hidden="true" class="border rounded-2 flex-shrink-0 color-bg-subtle
px-1 color-fg-muted ml-1 f6 d-none d-on-nav-focus js-jump-to-badge-jump">
  Jump to
  <span class="d-inline-block ml-1 v-align-middle">⌵</span>
</div>

```

```

  </div>
</label>

```

```

</div>

<div class="position-relative mr-lg-3 d-lg-inline-block">
  <a href="/login?return_to=https%3A%2F%2Fgithub.com%2Fhttps-deeplearning-
ai%2Fmachine-learning-engineering-for-production-
public%2Fblob%2Fmain%2Fcourse4%2Fweek1-ungraded-
labs%2FC4_W1_Lab_1_Docker_Intro.md"
    class="HeaderMenu-link HeaderMenu-link--sign-in flex-shrink-0 no-
underline d-block d-lg-inline-block border border-lg-0 rounded rounded-lg-0 p-2 p-
lg-0"
    data-hydro-click="{&quot;event_type&quot;:&quot;authentication.click&quot;;&quot;payload&quot;:
{&quot;location_in_page&quot;:&quot;site header
menu&quot;;&quot;repository_id&quot;:null,&quot;auth_type&quot;:&quot;SIGN_UP&quot;
;&quot;originating_url&quot;:&quot;https://github.com/https-deeplearning-
ai/machine-learning-engineering-for-production-public/blob/main/course4/week1-
ungraded-labs/C4_W1_Lab_1_Docker_Intro.md&quot;;&quot;user_id&quot;:null}}" data-
hydro-click-
hmac="301894dcfb24724596968b8ff5a37f42982035b7991040d92dc052b41bd8cbb6"
    data-ga-click="(Logged out) Header, clicked Sign in, text:sign-in">
    Sign in
  </a>
</div>

  <a href="/signup?
ref_cta=Sign+up&amp;ref_loc=header+logged+out&amp;ref_page=%2F%3Cuser-
name%3E%2F%3Crepo-name%3E%2Fblob%2Fshow&amp;source=header-
repo&amp;source_repo=https-deeplearning-ai%2Fmachine-learning-engineering-for-
production-public"
    class="HeaderMenu-link HeaderMenu-link--sign-up flex-shrink-0 d-none d-
lg-inline-block no-underline border color-border-default rounded px-2 py-1"
    data-hydro-click="{&quot;event_type&quot;:&quot;authentication.click&quot;;&quot;payload&quot;:
{&quot;location_in_page&quot;:&quot;site header
menu&quot;;&quot;repository_id&quot;:null,&quot;auth_type&quot;:&quot;SIGN_UP&quot;
;&quot;originating_url&quot;:&quot;https://github.com/https-deeplearning-
ai/machine-learning-engineering-for-production-public/blob/main/course4/week1-
ungraded-labs/C4_W1_Lab_1_Docker_Intro.md&quot;;&quot;user_id&quot;:null}}" data-
hydro-click-
hmac="301894dcfb24724596968b8ff5a37f42982035b7991040d92dc052b41bd8cbb6"
    data-analytics-event="{&quot;category&quot;:&quot;Sign
up&quot;;&quot;action&quot;:&quot;click to sign up for
account&quot;;&quot;label&quot;:&quot;ref_page:/&lt;user-name&gt;/&lt;repo-
name&gt;/blob/show;ref_cta:Sign up;ref_loc:header logged out&quot;}"
  >
    Sign up
  </a>
</div>
</div>
</div>

```

```
<div hidden="hidden" data-view-component="true" class="js-stale-session-flash flash flash-warn mb-3">
```

```
  <svg aria-hidden="true" height="16" viewBox="0 0 16 16" version="1.1" width="16" data-view-component="true" class="octicon octicon-alert">
    <path d="M6.457 1.047c.659-1.234 2.427-1.234 3.086 0l6.082 11.378A1.75 1.75 0 0 1 14.082 15H1.918a1.75 1.75 0 0 1-1.543-2.575Zm1.763.707a.25.25 0 0 0-.44 0L1.698 13.132a.25.25 0 0 0 .22.368h12.164a.25.25 0 0 0 .22-.368Zm.53 3.996v2.5a.75.75 0 0 1-1.5 0v-2.5a.75.75 0 0 1 1.5 0ZM9 11a1 1 0 1 1-2 0 1 1 0 0 1 2 0Z"></path>
```

```
<button class="flash-close js-flash-close" type="button" aria-label="Close">
  <svg aria-hidden="true" height="16" viewBox="0 0 16 16" version="1.1" width="16" data-view-component="true" class="octicon octicon-x">
    <path d="M3.72 3.72a.75.75 0 0 1 1.06 0L8 6.94l3.22-3.22a.749.749 0 0 1 1.275.326.749.749 0 0 1-.215.734L9.06 8l3.22 3.22a.749.749 0 0 1-.326 1.275.749.749 0 0 1-.734-.215L8 9.06l-3.22 3.22a.751.751 0 0 1-1.042-.018.751.751 0 0 1-.018-1.042L6.94 8 3.72 4.78a.75.75 0 0 1 0-1.06Z"></path>
```

```
<div id="js-flash-container" data-turbo-replace>
```

```
<include-fragment class="js-notification-shelf-include-fragment" data-base-src="https://github.com/notifications/beta/shelf"></include-fragment>
```

```
<div class="d-flex flex-wrap flex-justify-end mb-3 px-3 px-md-4 px-lg-5" style="gap: 1rem;">
```

```
  <div class="flex-auto min-width-0 width-fit mr-3">
```



```
<span class="author flex-self-stretch" itemprop="author">
  <a class="url fn" rel="author" data-hovercard-type="organization" data-hovercard-url="/orgs/https-deeplearning-ai/hovercard" data-octo-click="hovercard-link-click" data-octo-dimensions="link_type:self" href="/https-deeplearning-ai">
    https-deeplearning-ai
```

/ [machine-learning-engineering-for-production-public](#)

```
<span></span><span class="Label Label--secondary v-align-middle mr-1">Public</span>
```

```
</div>
```

```
<div id="repository-details-container" data-turbo-replace>
  <ul class="pagehead-actions flex-shrink-0 d-none d-md-inline" style="padding: 2px 0;">
```

 [Notifications](#)

 [Fork 1.9k](#)

 [Star 1.5k](#) 

```
</div>
</div>
```

```
<div id="responsive-meta-container" data-turbo-replace>
```

```
<nav data-pjax="#js-repo-pjax-container" aria-label="Repository" data-view-
component="true" class="js-repo-nav js-sidenav-container-pjax js-responsive-underlinenav
overflow-hidden UnderlineNav px-3 px-md-4 px-lg-5">
```

- ```
<svg aria-hidden="true" height="16" viewBox="0 0 16 16" version="1.1"
width="16" data-view-component="true" class="octicon octicon-code UnderlineNav-octicon
d-none d-sm-inline">
<path d="m11.28 3.22 4.25 4.25a.75.75 0 0 1 0 1.06l-4.25 4.25a.749.749 0 0 1-
1.275-.326.749.749 0 0 1 .215-.734l13.94 8l-3.72-3.72a.749.749 0 0 1 .326-1.275.749.749
0 0 1 .734.215zm-6.56 0a.751.751 0 0 1 1.042.018.751.751 0 0 1 .018 1.042l2.06 8l3.72
3.72a.749.749 0 0 1-.326 1.275.749.749 0 0 1-.734-.215l.47 8.53a.75.75 0 0 1 0-1.06z">
</path>
```

## [Code](#)

- ```
<svg aria-hidden="true" height="16" viewBox="0 0 16 16" version="1.1"
width="16" data-view-component="true" class="octicon octicon-issue-opened UnderlineNav-
octicon d-none d-sm-inline">
<path d="M8 9.5a1.5 1.5 0 1 0 0-3 1.5 1.5 0 0 0 0 3z"></path><path d="M8 0a8 8 0 1 1 0
16A8 8 0 0 1 8 0zM1.5 8a6.5 6.5 0 1 0 13 0 6.5 6.5 0 0 0-13 0z"></path>
```

[Issues 18](#)

- ```
<svg aria-hidden="true" height="16" viewBox="0 0 16 16" version="1.1"
width="16" data-view-component="true" class="octicon octicon-git-pull-request
UnderlineNav-octicon d-none d-sm-inline">
<path d="M1.5 3.25a2.25 2.25 0 1 1 3 2.122v5.256a2.251 2.251 0 1 1-1.5 0V5.372A2.25 2.25
0 0 1 1.5 3.25zm5.677-.177L9.573.677A.25.25 0 0 1 10 .854V2.5h1A2.5 2.5 0 0 1 13.5
5v5.628a2.251 2.251 0 1 1-1.5 0V5a1 1 0 0 0-1-1h-1v1.646a.25.25 0 0 1-.427.177L7.177
3.427a.25.25 0 0 1 0-.354zM3.75 2.5a.75.75 0 1 0 0 1.5.75.75 0 0 0 0-1.5zm0 9.5a.75.75 0
1 0 0 1.5.75.75 0 0 0 0-1.5zm8.25.75a.75.75 0 1 0 1.5 0 .75.75 0 0 0-1.5 0z"></path>
```

## [Pull requests 2](#)

- ```

<svg aria-hidden="true" height="16" viewBox="0 0 16 16" version="1.1"
width="16" data-view-component="true" class="octicon octicon-play UnderlineNav-octicon
d-none d-sm-inline">
<path d="M8 0a8 8 0 1 1 0 16A8 8 0 0 1 8 0ZM1.5 8a6.5 6.5 0 1 0 13 0 6.5 6.5 0 0 0-13
0Zm4.879-2.773 4.264 2.559a.25.25 0 0 1 0 .428l-4.264 2.559A.25.25 0 0 1 6
10.559V5.442a.25.25 0 0 1 .379-.215Z"></path>

```

[Actions](#)

- ```

<svg aria-hidden="true" height="16" viewBox="0 0 16 16" version="1.1"
width="16" data-view-component="true" class="octicon octicon-table UnderlineNav-octicon
d-none d-sm-inline">
<path d="M0 1.75C0 .784.784 0 1.75 0h12.5C15.216 0 16 .784 16 1.75v12.5A1.75 1.75 0 0 1
14.25 16H1.75A1.75 1.75 0 0 1 0 14.25ZM6.5 6.5v8h7.75a.25.25 0 0 0 .25-.25V6.5Zm8-
1.5V1.75a.25.25 0 0 0-.25-.25H6.5V5Zm-13 1.5v7.75c0 .138.112.25.25.25H5v-8ZM5
5V1.5H1.75a.25.25 0 0 0-.25.25V5Z"></path>

```

### [Projects](#)

- ```

<svg aria-hidden="true" height="16" viewBox="0 0 16 16" version="1.1"
width="16" data-view-component="true" class="octicon octicon-shield UnderlineNav-octicon
d-none d-sm-inline">
<path d="M7.467.133a1.748 1.748 0 0 1 1.066 0l5.25 1.68A1.75 1.75 0 0 1 15 3.48V7c0
1.566-.32 3.182-1.303 4.682-.983 1.498-2.585 2.813-5.032 3.855a1.697 1.697 0 0 1-1.33
0c-2.447-1.042-4.049-2.357-5.032-3.855C1.32 10.182 1 8.566 1 7V3.48a1.75 1.75 0 0 1
1.217-1.667Zm.61 1.429a.25.25 0 0 0-.153 0l-5.25 1.68a.25.25 0 0 0-.174.238V7c0
1.358.275 2.666 1.057 3.86.784 1.194 2.121 2.34 4.366 3.297a.196.196 0 0 0 .154
0c2.245-.956 3.582-2.104 4.366-3.298C13.225 9.666 13.5 8.36 13.5 7V3.48a.251.251 0 0
0-.174-.237l-5.25-1.68ZM8.75 4.75v3a.75.75 0 0 1-1.5 0v-3a.75.75 0 0 1 1.5 0ZM9 10.5a1 1
0 1 1-2 0 1 1 0 0 1 2 0Z"></path>

```

[Security](#)

- ```

<svg aria-hidden="true" height="16" viewBox="0 0 16 16" version="1.1"
width="16" data-view-component="true" class="octicon octicon-graph UnderlineNav-octicon
d-none d-sm-inline">
<path d="M1.5 1.75V13.5h13.75a.75.75 0 0 1 0 1.5H.75a.75.75 0 0 1-.75-.75V1.75a.75.75 0
0 1 1.5 0Zm14.28 2.53-5.25 5.25a.75.75 0 0 1-1.06 0L7 7.06 4.28 9.78a.751.751 0 0 1-
1.042-.018.751.751 0 0 1-.018-1.042l3.25-3.25a.75.75 0 0 1 1.06 0L10 7.94l4.72-
4.72a.751.751 0 0 1 1.042.018.751.751 0 0 1 .018 1.042Z"></path>

```

### [Insights](#)

### [Permalink](#)

► 🔑 main



[View all tags](#)

## Name already in use

✕

```
<div data-view-component="true"> A tag already exists with the provided branch
name. Many Git commands accept both tag and branch names, so creating this branch may cause
unexpected behavior. Are you sure you want to create this branch?
```

```
</div>
<footer class="Overlay-footer Overlay-footer--alignEnd">
 <button data-close-dialog-id="warn-tag-match-create-branch-dialog" type="button"
data-view-component="true" class="btn"> Cancel
```

Create

### [machine-learning-engineering-for-production-public/course4/week1-ungraded-labs/C4\\_W1\\_Lab\\_1\\_Docker\\_Intro.md](#)

[Go to file](#)

► ...

```
<div id="spoof-warning" class="mt-0 pb-3" hidden aria-hidden>
```

```
<svg aria-hidden="true" height="16" viewBox="0 0 16 16" version="1.1" width="16" data-view-
component="true" class="octicon octicon-alert float-left mt-1">
<path d="M6.457 1.047c.659-1.234 2.427-1.234 3.086 0l6.082 11.378A1.75 1.75 0 0 1 14.082
15H1.918a1.75 1.75 0 0 1-1.543-2.575Zm1.763.707a.25.25 0 0 0-.44 0L1.698 13.132a.25.25 0 0
0 .22.368h12.164a.25.25 0 0 0 .22-.368Zm.53 3.996v2.5a.75.75 0 0 1-1.5 0v-2.5a.75.75 0 0 1
1.5 0ZM9 11a1 1 0 1 1-2 0 1 1 0 0 1 2 0Z"></path>
```

```
<div class="overflow-hidden">This commit does not belong to any branch on this repository,
and may belong to a fork outside of the repository.</div>
```

```
<include-fragment src="/https-deeplearning-ai/machine-learning-engineering-for-production-
public/spoofed_commit_check/34af5af3f5b09a52a1537292b0fe14a29ae16b8d" data-test-
selector="spoofed-commit-check"></include-fragment>
```

```
<div class="Box d-flex flex-column flex-shrink-0 mb-3">
```

```
<div class="Box-body d-flex flex-items-center" >
 <div class="Skeleton Skeleton--text col-1"> </div>
 Cannot retrieve contributors at this
time
</div>
```

```
<readme-toc>
```

```
<div data-target="readme-toc.content" class="Box mt-3 position-relative">
```

```
<details
```

```
data-target="readme-toc.trigger" data-menu-hydro-
click="{\"event_type\":\"repository_toc_menu.click\",\"payload\":
{\"target\":\"trigger\",\"repository_id\":313476213,\"originating_url\":\"https://github.com/https-deeplearning-
ai/machine-learning-engineering-for-production-public/blob/main/course4/week1-ungraded-
labs/C4_W1_Lab_1_Docker_Intro.md\"""user_id":null}}\" data-menu-hydro-click-
hmac=\"a311f902a183dc5c9768f33e0d7617bb4801ec42e6559b025ae4fda6288c1086\" class=\"dropdown
details-reset details-overlay\"
```



```

<div class="SelectMenu-filter">
 <input
 class="SelectMenu-input form-control js-filterable-field"
 id="toc-filter-field"
 type="text"
 autocomplete="off"
 spellcheck="false"
 autofocus
 placeholder="Filter headings"
 aria-label="Filter headings">
</div>

```

```

<div class="SelectMenu-list SelectMenu-list--borderless p-2" style="overscroll-behavior:
contain;" data-filterable-for="toc-filter-field" data-filterable-type="substring">
 <a role="menuitem" class="filter-item SelectMenu-item ws-normal wb-break-word line-
clamp-2 py-1 text-emphasized" style="-webkit-box-orient: vertical; padding-left: 12px;"
data-action="click:readme-toc#blur" data-targets="readme-toc.entries" data-hydro-click="{
'event_type': 'repository_toc_menu.click', 'payload': {
'target': 'entry', 'repository_id': '313476213', 'originating_
url': 'https://github.com/https-deeplearning-ai/machine-learning-engineering-for-
production-public/blob/main/course4/week1-ungraded-
labs/C4_W1_Lab_1_Docker_Intro.md', 'user_id': null }}" data-hydro-click-
hmac="eb8f869b3e7d7dcc3db61fedaa6968fe6c2180f8fa0784dab65d861469a2d702" href="#ungraded-
labs-intro-to-docker-and-installation">Ungraded Labs: Intro to Docker and installation
 <a role="menuitem" class="filter-item SelectMenu-item ws-normal wb-break-word line-
clamp-2 py-1 " style="-webkit-box-orient: vertical; padding-left: 24px;" data-
action="click:readme-toc#blur" data-targets="readme-toc.entries" data-hydro-click="{
'event_type': 'repository_toc_menu.click', 'payload': {
'target': 'entry', 'repository_id': '313476213', 'originating_
url': 'https://github.com/https-deeplearning-ai/machine-learning-engineering-for-
production-public/blob/main/course4/week1-ungraded-
labs/C4_W1_Lab_1_Docker_Intro.md', 'user_id': null }}" data-hydro-click-
hmac="eb8f869b3e7d7dcc3db61fedaa6968fe6c2180f8fa0784dab65d861469a2d702"
href="#docker">Docker
 <a role="menuitem" class="filter-item SelectMenu-item ws-normal wb-break-word line-
clamp-2 py-1 " style="-webkit-box-orient: vertical; padding-left: 36px;" data-
action="click:readme-toc#blur" data-targets="readme-toc.entries" data-hydro-click="{
'event_type': 'repository_toc_menu.click', 'payload': {
'target': 'entry', 'repository_id': '313476213', 'originating_
url': 'https://github.com/https-deeplearning-ai/machine-learning-engineering-for-
production-public/blob/main/course4/week1-ungraded-
labs/C4_W1_Lab_1_Docker_Intro.md', 'user_id': null }}" data-hydro-click-
hmac="eb8f869b3e7d7dcc3db61fedaa6968fe6c2180f8fa0784dab65d861469a2d702" href="#why-
docker">Why Docker?
 <a role="menuitem" class="filter-item SelectMenu-item ws-normal wb-break-word line-
clamp-2 py-1 " style="-webkit-box-orient: vertical; padding-left: 36px;" data-
action="click:readme-toc#blur" data-targets="readme-toc.entries" data-hydro-click="{
'event_type': 'repository_toc_menu.click', 'payload': {
'target': 'entry', 'repository_id': '313476213', 'originating_
url': 'https://github.com/https-deeplearning-ai/machine-learning-engineering-for-
production-public/blob/main/course4/week1-ungraded-
labs/C4_W1_Lab_1_Docker_Intro.md', 'user_id': null }}" data-hydro-click-
hmac="eb8f869b3e7d7dcc3db61fedaa6968fe6c2180f8fa0784dab65d861469a2d702" href="#some-key-

```

concepts">Some key concepts</a>

</div>

</div>

78 lines (48 sloc)

<span class="file-info-divider"></span>

5.66 KB

**most useful when they interact with the outside world.** There is a knowledge gap from successfully training a model in a local environment and actually serving it. That is what these ungraded labs are for.

```
<div class="BtnGroup">
 <a data-permalink-href="/https-deeplearning-ai/machine-learning-engineering-for-
production-public/raw/34af5af3f5b09a52a1537292b0fe14a29ae16b8d/course4/week1-ungraded-
labs/C4_W1_Lab_1_Docker_Intro.md" href="/https-deeplearning-ai/machine-learning-
engineering-for-production-public/raw/main/course4/week1-ungraded-
labs/C4_W1_Lab_1_Docker_Intro.md" id="raw-url" group_item="true" data-view-component="true"
class="js-permalink-replaceable-link Button--secondary Button--small Button">
 Raw

```

[interpreters](#) or [compilers](#) for it to run.

```
<div class="d-flex">
```

 For this you created a virtual environment in your local machine. Everything works fine but now you want to share this model with a colleague of yours who does not have Python installed, much less any of the required libraries.

```
<button class="btn-octicon btn-octicon-danger disabled tooltiped tooltiped-nw"
disabled
 aria-label="You must be signed in to make or propose changes" type="button">
 <svg aria-hidden="true" height="16" viewBox="0 0 16 16" version="1.1" width="16"
data-view-component="true" class="octicon octicon-trash">
<path d="M11 1.75V3h2.25a.75.75 0 0 1 0 1.5H2.75a.75.75 0 0 1 0-1.5H5V1.75C5 .784 5.784 0
6.75 0h2.5C10.216 0 11 .784 11 1.75ZM4.496 6.675l.66 6.6a.25.25 0 0 0 .249.225h5.19a.25.25
0 0 0 .249-.225l.66-6.6a.75.75 0 0 1 1.492.149l-.66 6.6A1.748 1.748 0 0 1 10.595 15h-
5.19a1.75 1.75 0 0 1-1.741-1.575l-.66-6.6a.75.75 0 1 1 1.492-.15ZM6.5 1.75V3h3V1.75a.25.25
0 0 0-.25-.25h-2.5a.25.25 0 0 0-.25.25Z"></path>
```

- **Dockerfile** : This is a special file that contains all of the instructions required to build an image. These instructions can be anything from "install Python version 3.7" to "copy my code inside the image".
- **Image** : This refers to the collection of all your software in one single place. Using the previous example, the image will include Python, Tensorflow, JAX and your code. This will be achieved by setting the appropriate instructions within the Dockerfile.
- **Container** : This a running instance of an image. Images by themselves don't do much aside from saving the information of your code and its dependencies. You need to run a container out of them to actually run the code within. Containers are usually meant to perform a single task but they can be used as runtimes to run software that you haven't installed.

Now that you have a high level idea of how Docker works it is time for you to install it. If you already have it installed you can skip most of the following items.

## ■ ↗ Installation

---

To install the free version of Docker visit this [link](#).

## ■ ↗ Note for Windows users

---

If you are currently on a Windows machine we recommend you install [WSL2](#), in particular a Debian-based distribution such as **Ubuntu**. This will enable a real Linux kernel inside your Windows OS and will get full usage out of Docker and other command line tools you will be using such as [curl](#). **Machine learning deployment is usually done in Linux machines on the cloud so it is good that you get to know this OS for the purpose of deploying your models.**

If you already have WSL2 installed check out this [link](#) to complete the Docker Desktop installation.

**All of the local ungraded labs in this course should be run using a WSL2 shell.** To open such shell use the Windows search bar and type either `wsl` or `bash`, one of these should be available if you installed WSL2 previously.

You may also like to install the [Windows Terminal](#) since it allows for a better experience to deal with all your shells and command lines.

---

## ■ ↗ Testing the Docker installation

---

To test your installation go into the command line and type the following command:

```
docker run --rm docker/whalesay cowsay "deploying ML models is fun"
```

This command will try to run the `docker/whalesay` image. Since you probably don't have this image installed, Docker will look for it in the [Docker Hub](#) and pull it to your machine to run it. If everything worked well you should see the following image printed in your command line:



If you think of a funny thing for the Docker whale to say be sure to replace the message above with it and send the image to your friends.

If the command didn't work try following the previous steps again.

---

## ↗ Check your curl installation

Another tool that you will be using during these labs is [curl](#) so be sure to check that it is installed. To do so use the following command:

```
curl -V
```

If the `curl` command is not recognized then you can install it on a Debian-based Linux (like Ubuntu) machine with this command:

```
apt-get install curl
```

If you are on other OS like Mac, look up on google how to install it, it should be pretty straightforward.

`curl` is just an HTTP client, if you prefer other or an app with a graphical interface instead of the command line you can use something like [postman](#).

Now you are ready to complete the rest of ungraded labs in this course! **Nice job!**

```
</div>
```



```
</main>
```

```
<footer class="footer width-full container-xl p-responsive" role="contentinfo">
```

## ■ Footer



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```

<nav aria-label='Footer' class="col-12 col-lg-8">
 <h3 class='sr-only' id='sr-footer-heading'>Footer navigation</h3>
 <ul class="list-style-none d-flex flex-wrap col-12 flex-justify-center flex-lg-justify-between mb-2 mb-lg-0" aria-labelledby='sr-footer-heading'>
 <li class="mr-3 mr-lg-0">Terms
 <li class="mr-3 mr-lg-0">Privacy
 <li class="mr-3 mr-lg-0"><a data-analytics-event="{"category":"Footer", "action":"go to security", "label":"text:security"}" href="https://github.com/security">Security
 <li class="mr-3 mr-lg-0">Status
 <li class="mr-3 mr-lg-0"><a data-ga-click="Footer, go to help, text:Docs" href="https://docs.github.com">Docs
 <li class="mr-3 mr-lg-0">Contact GitHub
 <li class="mr-3 mr-lg-0">Pricing
 <li class="mr-3 mr-lg-0">API
 <li class="mr-3 mr-lg-0">Training
 <li class="mr-3 mr-lg-0">Blog
 <a data-ga-click="Footer, go to about, text:about" href="https://github.com/about">About

</nav>

```

```
<template id="site-details-dialog">
```



```

<div class="Popover js-hovercard-content position-absolute" style="display: none; outline: none;" tabindex="0">

```



```
<template id="snippet-clipboard-copy-button">
```



```
</div>
```

```
<div id="js-global-screen-reader-notice" class="sr-only" aria-live="polite" ></div>
```